

Climatic influences on forest fire and mountain pine beetle outbreaks and resulting runoff effects in large watersheds in British Columbia, Canada

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Abstract

Many studies have defined the interrelationships between climate, forest disturbance, and runoff at small scales (<100 km²), but few have translated these relationships to large watersheds (>500 km²). In this study, we explore the relationship between climate, extreme forest fire seasons, mountain pine beetle (MPB) outbreaks, and runoff in eight large watersheds within the Fraser and Peace drainage basins in British Columbia (BC), Canada from 1981–2019. Using a climate index based on precipitation and air temperature anomalies, we find extreme forest fire seasons (those that burned >5% of a watershed's area) are most likely to occur when a warm/dry summer is preceded by multiple seasons of cool/wet conditions. Using the climate suitability class (CSC) model to explore the relationship between climate and MPB outbreaks, we validate previous findings that lower-than-average precipitation, warm growing season temperatures, and lack of extremely cold temperatures during winter are connected to MPB outbreaks within central BC. However, the CSC model needs improvements to accurately assess MPB suitability in northern watersheds that are located outside the model's calibration region, either through weighted variables or lower degree day thresholds. Minimal runoff response occurs from these forest disturbances, with the most prominent runoff change being related to the 2014 fire season in the Osilinka and Mesilinka watersheds. The limited effects of forest disturbance on annual runoff are likely related to large watershed sizes, low percentages of disturbed area in some study watersheds and post-MPB forest dynamics. These results provide valuable insight into the interrelationships of climate, forest disturbance and runoff in large Canadian boreal forested watersheds.

KEYWORDS

British Columbia, climate, forest disturbance, forest fires, mountain pine beetle, runoff

1 | INTRODUCTION

Both forest disturbance and climatic variability influence the hydrologic regime throughout western North America (Owens et al., 2013; Winkler, Spittlehouse, & Boon, 2017; Winkler, Spittlehouse, Boon, & Zimonick, 2015; Zhang & Wei, 2012). Within British Columbia (BC),

Canada, forest resources, which cover nearly two thirds of the province, are experiencing significant changes from insect infestations (mountain pine beetle (MPB), spruce bark beetle, tent caterpillars, etc.), wildfires, and timber harvesting. MPB outbreaks have affected nearly 183,000 km² of BC since the early 1990's (Corbett, Withey, Lantz, & Ochuodho, 2016) and extreme fire events have burned

unprecedented amounts of forest, with over 13,500 km² of BC burning in 2018 alone (Government of British Columbia, 2019c). Concurrently, changes in BC's climate regime are occurring. The Government of British Columbia reports average annual air temperatures increasing by 1.4°C per century since 1900, with winter temperatures warming by an average of 2.2°C per century (British Columbia Ministry of Environment, 2015). While seasonal and annual precipitation are highly variable across BC, it is projected that more intense and frequent heavy-rain events will occur while droughts and water shortages increase (British Columbia Ministry of Environment, 2015).

On forest-stand or small-basin scales, a clear interrelationship between climate, forest disturbance, and the hydrologic regime exists. Increasing winter temperatures decrease snowfall (Rasouli, Pomeroy, & Whitfield, 2019), advance spring melt (Islam, Déry, & Werner, 2017), and prevents MPB broods from dying off due to the lack of prolonged cold spells (Safranyik & Linton, 1998). Increasing summer temperatures and decreasing precipitation can lead to changes in evapotranspiration rates (Rasouli et al., 2019) and increase forest fire activity. As forest disturbance escalates, a reduction in canopy cover decreases interception (Owens et al., 2013; Winkler et al., 2015; Winkler, Boon, Zimonick, & Spittlehouse, 2014), suppressing evapotranspiration rates, increasing solar radiation near the forest floor, and changing forest floor characteristics (soil infiltration, soil cohesiveness, etc.) (Owens et al., 2013; Zhang & Wei, 2012). These changes, in turn, lead to a disruption in the surface water budget, such as streamflow and runoff regimes (Winkler et al., 2015, 2017). The climate can also be affected by large scale forest disturbance through alterations in surface heat flux, albedo, and surface air temperatures after large forest fires and MPB attacks (Amiro, MacPherson, & Desjardins, 1999; Maness, Kushner, & Fung, 2012).

There are several case studies of large (>500 km²) watersheds in BC that investigate the response of streamflow to forest disturbance (MPB, logging and forest fires) and climate change. In all case studies, offsetting of climatic variability and forest disturbance effects on annual mean streamflow is found (Li et al., 2018; Wei & Zhang, 2010; Zhang & Wei, 2012, 2014; Zhang, Wei, & Li, 2017), however the influence of climate on forest disturbance is not addressed. Differences in streamflow magnitude change exist between these case studies likely due to the heterogeneity of climate and watershed properties such as topography, soils, vegetation types, and geology.

This study focuses on the effects annual and seasonal precipitation and air temperature have on the initiation of extreme forest fire seasons and MPB outbreaks and the resulting changes to runoff in large BC watersheds. Warmer and drier climates tend to increase fuel supply for forest fires, allowing them to more easily spread (Littell, McKenzie, Peterson, & Westerling, 2009). At forest-stand or small watershed scales, many studies report decreased soil infiltration rates as a result of increasing hydrophobic soils post-fire leading to increased overland flow, with magnitudes of change dependent on fire severity (Moody et al., 2016; Moreno, Gourley, Pham, & Spade, 2020; Shakesby & Doerr, 2006; Stoof et al., 2012). This increase in overland flow, especially in the wake of large rain events, is linked to increasing maximum flows.

Like forest fires, MPB outbreaks are also influenced by air temperature and precipitation. Annual and seasonal temperatures often dictate the success of MPB broods as well as their ability to synchronize attacks on tree stands (Carroll, Tylor, Régnière, & Safranyik, 2004; Logan & Powell, 2001; Safranyik & Linton, 1998; Safranyik, Shrimpton, & Whitney, 1975; Taylor, Carroll, Alfaro, & Safranyik, 2006), while precipitation influences the host tree's ability to prevent infestations from occurring (Berg, Henry, Fastie, De Volder, & Matsuoka, 2006; Raffa et al., 2008). Unlike forest fires, MPB attacks mostly affect the forest canopy (Alfaro, van Akker, & Hawkes, 2015), leaving much of the understory unchanged. The overstory is affected in stages after an infestation, with needles gradually changing from green to red before falling off, usually within 3–5 years post-attack, increasing needle litter on the forest floor and decreasing precipitation interception (Winkler et al., 2014). Minimal changes to streamflow following MPB outbreaks in western North America have been found in several studies (Biederman et al., 2015; Slinski, Hogue, Porter, & Mccray, 2016), with Slinski et al. (2016) reporting the most significant changes in smaller watersheds and those with high spring precipitation.

In this study, we use normalized precipitation and air temperature anomalies and historical forest fire records to analyse the relationships between extreme forest fire seasons (those that burned >5% of a watershed's area) and climate. Flow records pre- and post-fire seasons are examined to understand the annual and seasonal hydrological changes at large watershed scales. A climate suitability class (CSC) model (Carroll et al., 2004; Safranyik et al., 1975; Taylor et al., 2006) is used in tandem with MPB equivalent clear-cut area (ECA) timeseries to explore precipitation and air temperature trends that lead to MPB outbreaks. Streamflow records are compared over outbreak periods to explore possible changes in annual runoff.

2 | STUDY SITES

The Fraser and Peace River basins in BC are important ecologically and economically within the province and beyond. The Fraser River is the longest river in BC, flowing from Mount Robson Provincial Park near Jasper, Alberta to the Pacific Ocean in Vancouver. The Fraser River is a critical salmon habitat, sustaining one of the largest migrations of Pacific Ocean sockeye salmon in the world (Beacham et al., 2005). The Peace River, which flows from north-central BC to the Slave River in northeastern Alberta, powers the WAC Bennett Hydroelectric Dam, Peace Canyon Dam, and Site C Dam and Reservoir (currently in development) in BC. The Peace River is also a main contributor to the largest boreal inland delta in the world, the Peace-Athabasca Delta in northeastern Alberta, which is a Ramsar listed wetland region that sustains large migratory bird and fish populations (Timoney, 2013).

We explore eight large, unregulated sub-watersheds that are representative of forest disturbance and climate change in the Fraser and Peace River basins to understand the interrelationships occurring in these regions. The Spius, Chilko, Chilcotin, Stellako and Nautley

watersheds lie in the Fraser drainage basin in south-central BC and the Nation, Osilinka, and Mesilinka watersheds lie in the Peace drainage basin in north-central BC (Figure 1). These watersheds cover a $\sim 7^\circ \times \sim 7^\circ$ area within central BC and have extended natural flow records (≥ 40 years), high levels of forest disturbance from MPB outbreaks and forest fires, and snow-dominated hydrologic regimes.

The Spius, Chilko and Chilcotin are the most southern watersheds in the study region (Figure 1). The smallest, Spius Creek (777 km²), spans several biogeoclimatic zones with the most dominant being Engelmann Spruce-Subalpine Fir (46%), with 27% being covered in pine tree species. MPB ECA has rapidly increased since 2004 in the Spius, with 17% of the watershed affected by 2018 (Table 1). The Chilko watershed (6,876 km²), which forms the southwestern portion of the Chilcotin (19,206 km²), contains the highest elevations, with

23% of the Chilko's area lying above 2,000 m (Figure 1c, Table 1). The Chilko and Chilcotin's most dominant biogeoclimatic zone is Sub-Boreal Pine-Spruce, which covers 29 and 40% of the watersheds, respectively, with over 40% of the watershed's areas being covered by pine tree species.

Both the Chilko and Chilcotin are highly affected by MPB and forest fires, resulting in 30 and 45% cumulative fire and MPB ECA by 2018, respectively. The Chilcotin Plateau was severely affected by the 2000's MPB outbreak in BC (Alfaro et al., 2015) with MPB ECA increasing within the Chilcotin watershed by an average of 2.6% annually from 2004–2012. Large fires in 2009 and 2017 within the Chilko and Chilcotin, respectively, also drastically increase forest disturbance in these regions, increasing fire ECA by more than 10% in one year (Figure S1 in Data S1).

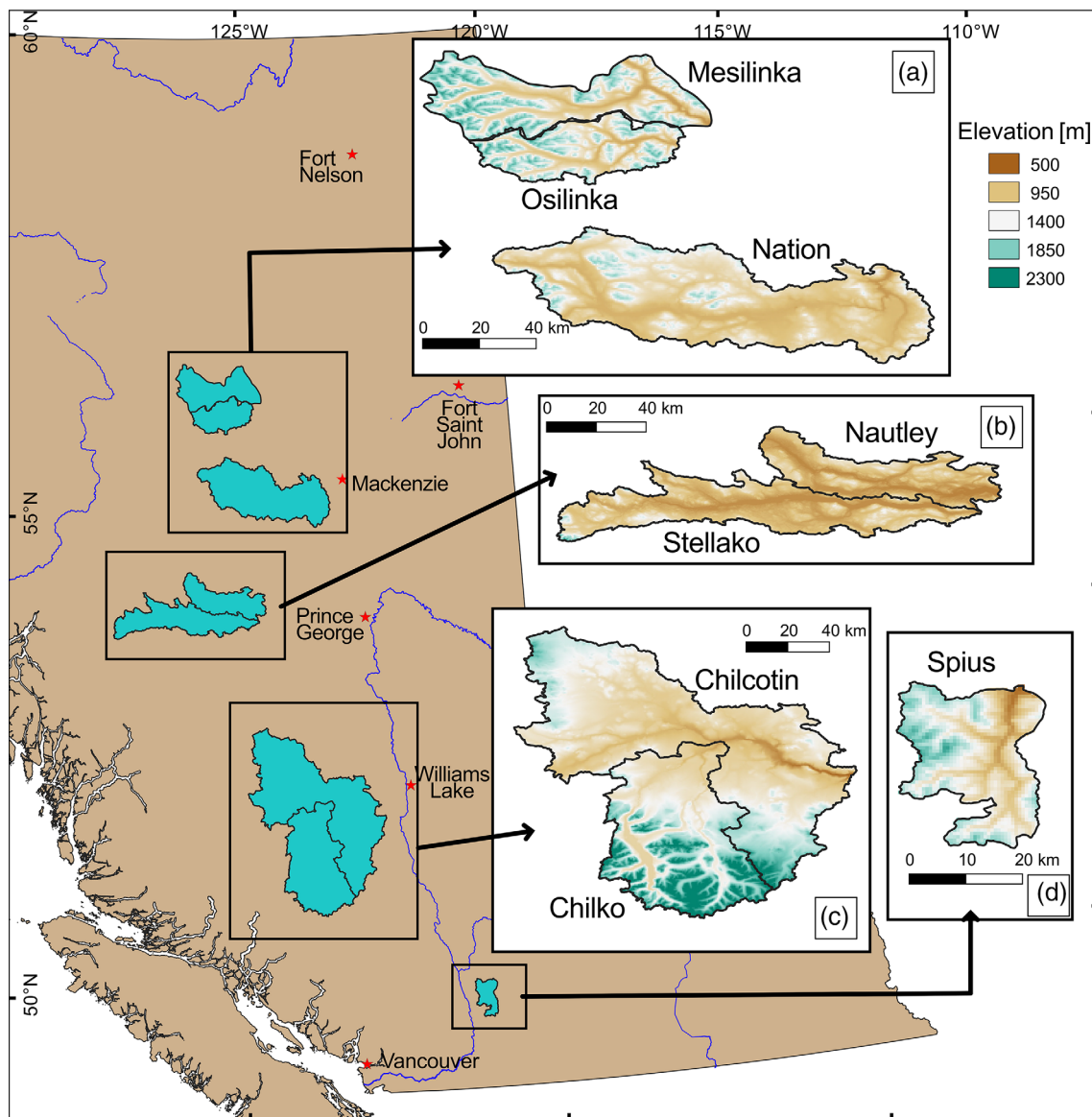


FIGURE 1 The eight watersheds (teal) within BC with subsets a-d depicting their elevations. (a) Mesilinka, Osilinka, and Nation watersheds; (b) Nautley and Stellako watersheds (the Stellako being a sub-basin of the Nautley); (c) Chilcotin and Chilko watersheds (the Chilko being a sub-basin of the Chilcotin); (d) Spius watershed

TABLE 1 Summary of physical features at the eight study sites

Watershed	Area [km ²]	Mean elevation [m]	Biogeoclimatic zones [%]								MPB ECA [%]	Forest fire ECA [%]	Pine cover [%]
			Alpine	BWBS	ESSF	IDF	MS	SBPS	SBS	SWB			
Spilus	775	1,493	0.6	—	45.9	27.4	25.4	—	—	—	17.9	4.2	24
Chilko	6,880	1,756	19.0	—	24.3	14.1	13.7	28.6	—	—	19.0	10.7	45
Chilcotin	19,205	1,528	7.6	—	11.8	21.7	18.5	40.1	—	—	27.3	17.8	52
Nation	6,790	1,258	0.2	—	44.1	—	—	—	55.7	—	15.2	0.8	23
Osilinka	1,950	1,519	6.7	14.3	78.9	—	—	—	—	—	12.6	8.7	17
Mesilinka	3,060	1,537	11.7	15.6	59.7	—	—	—	4.2	8.7	11.8	12.2	17
Stellako	4,020	969	0.8	—	11.5	—	—	—	87.8	—	9.2	25.7	26
Nautley	6,552	950	0.5	—	9.5	—	—	—	90.0	—	8.8	25.1	27

Note: The eight major biogeoclimatic zones represented are: (1) Alpine (sum of Boreal Altai Fescue Alpine, Coastal Mountain-heather Alpine, and Interior Mountain-heather Alpine); (2) Boreal White and Black Spruce (BWBS); (3) Engelmann Spruce-Subalpine Fir (ESSF); (4) Interior Douglas Fir (IDF); (5) Montane Spruce (MS); (6) Sub-Boreal Pine Spruce (7); Sub-Boreal Spruce (SBS); and (8) Spruce/Willow/Birch (SWB) (Government of British Columbia, 2019a). Percentage of equivalent clear-cut area (ECA) from forest fires and mountain pine beetle (MPB) representative of 2018 and percent of pine cover representative of 2019.

The Nautley watershed (6,552 km²) and its sub-basin, the Stellako watershed (4,020 km²), are also part of the Fraser River basin, via its tributary, the Nechako River. These watersheds have the lowest average elevation of all the study sites, at <1,000 m above sea level. Both watersheds are ~90% dominated by Sub-Boreal Spruce biogeoclimatic zones, with ~26% of the watersheds being covered by pine tree species. While MPB is not as prevalent in these watersheds as the other study sites, forest fires have affected over 25% of these watersheds, with a large forest fire in 2018 being the main contributor to the high forest fire ECA (Figure S1 in Data S1).

The other three watersheds, Nation (6,790 km²), Osilinka (1,928 km²), and Mesilinka (2,988 km²), lie in the north-central interior of BC (Figure 1a, Table 1). The Nation watershed spans several biogeoclimatic zones, much of which lies in Sub-Boreal Spruce (56%) and Engelmann Spruce-Subalpine Fir (44%), with pine tree species covering 23% of the watershed. Osilinka and Mesilinka are both predominantly covered by Engelmann Spruce-Subalpine Fir, at 79 and 60% respectively, with 17% of the watersheds being covered by pine trees. MPB ECA has significantly increased since 2004 within all Peace River sub-watersheds, with increases of about 1% annually. Large fires in the Osilinka and Mesilinka during 2014 caused fire ECA to increase by 6 and 12%, respectively, in a single year (Figure S1 in Data S1).

3 | DATA

The newly released ERA5-land gridded climate product from the European Centre for Medium-Range Weather Forecasts (ECMWF) is used to quantify precipitation and air temperature in all eight watersheds (Copernicus Climate Change Service (C3S), 2019). ERA5-land is a reanalysis product that applies the laws of physics to create a dataset based on modelled data and ground observations, with in-situ measurements influencing the model through the atmospheric forcing

input in simulations. ERA5-land estimates over 50 climatic variables from 1981-present at ~9 km resolution, with hourly 2m air temperature and total precipitation variables being used for this study. The spatially averaged monthly precipitation (air temperature) for each watershed is calculated by summing (averaging) the hourly totals (temperatures) by month at the grid cell scale then calculating the average of all grid cells that fall within a watershed's boundary.

Annual ECA (Giles-Hansen, Li, & Wei, 2019; Li et al., 2018; Wei & Zhang, 2010) is used to represent forest disturbance levels within the eight study watersheds. ECA is the percent of area that has been affected by forest fires, logging, or MPB while accounting for hydrologic recovery due to forest regeneration. ECA coefficients range from 0–100%, with 0% representing no disturbance or full recovery and 100% indicating complete removal of vegetation, with the concept of ECA having been successfully applied in BC (Hudson, 2000; Huggard, 2011; Zhang & Wei, 2014) and elsewhere (Ager & Clifton, 2005; Bettinger, Johnson, & Sessions, 1998).

MPB, forest fires, and logging ECA have different coefficients as each type of disturbance has different severities and consequently various hydrological impacts. For example, clear-cut causes removal of all trees, while MPB often leads to partial mortality of the overstory, leaving the understory vegetation minimally disturbed (Bearup, Maxwell, Clow, & McCray, 2014). As such, different ECA coefficients are assigned to reflect those differences between clear-cuts and MPB (Lewis & Huggard, 2010; Zhang & Wei, 2014). The ECA coefficient time series for fire and logging are based on tree species and their attributes, while coefficients for MPB are based on the severities of infested forest stands in different biogeoclimatic zones. In this study, the Vegetation Resource Inventory (VRI) (Government of British Columbia, 2020c) and the Consolidated Cutblock databases (Government of British Columbia, 2020b) from the BC Ministry of Forests, Lands, Natural Resource Operations and Rural Development (FLNRORD) are combined in ArcGIS and the three types of disturbances are extracted, with their sum equating to total cumulative

TABLE 2 Hydrometric stations used to obtain daily runoff data for the eight study watersheds (Environment and Climate Change Canada, 2020)

Name	ID	Data range
Spilus Creek near Canford	08LG008	1911–2019 ^a
Chilko River near Redstone	08MA001	1927–2019
Chilcotin River near Big Creek	08MB005	1970–2019
Nation River near the mouth	07ED003	1981–2019
Osilinka River near End Lake	07EC004	1981–2019
Mesilinka River at Gopherhole Creek	07EC003	1976–2019
Stellako River at Glenannan	08JB002	1929–2019
Nautley River near Fort Fraser	08JB003	1950–2019

^aMissing data from 2008–2011.

ECA. The ECA dataset is primarily used for identifying MPB outbreaks in each watershed.

The VRI database (Government of British Columbia, 2020c) is also used to determine the percent of pine tree cover within each watershed. The VRI stores the percent cover of up to six leading tree species within each polygon, of which we extract all pine species. The specific tree species change for each VRI polygon but the majority of pine trees in the region are lodgepole pines. We extract and sum the percent of pine species within each VRI polygon that is encompassed by the watershed's boundary and calculate how much of all the polygons within the watershed contain pine species. This provides us with an estimate of the total percent of pine within a watershed. While this process can be performed for other tree species, we are most interested in host trees of MPB, thus we only estimate pine tree distributions within this study.

The sizes of forest fire events are identified using the Historical Fire Perimeter database (Government of British Columbia, 2019b) and the Fire Burn Severity Database (Government of British Columbia, 2020a), provided by FLNRORD. The burn severity data are clipped to fire perimeter extents to categorize the burn severity within each fire perimeter as either unburned, low, medium, or high. We subtract areas identified as unburned within the fire perimeter from the total forest fire area to determine the area burned within a watershed during a given year.

Hydrological data are obtained from the Water Survey of Canada's HYDAT data archive (Environment and Climate Change Canada, 2020) with watershed runoff being determined from a hydrometric gauge located at or near the mouth of the river's basin (Table 2). All hydrometric gauge sites record the unregulated daily flow in $\text{m}^3 \text{s}^{-1}$ which is converted to km^3/month . Assuming that flow is constant throughout the watershed, the monthly discharge is divided by watershed area to produce average monthly runoff in mm.

4 | METHODS

4.1 | Climate index

To explore the relationship between forest fires and climate, we use a unitless climate index, CI , based on the normalized precipitation

anomaly (NPA) and normalized temperature anomaly (NTA) for a given year or season (Barber, Juday, & Finney, 2000; Déry, Stieglitz, Rennermalm, & Wood, 2005), which are defined as follows:

$$NPA_i = \frac{P_i - \bar{P}}{S_P}, NTA_i = \frac{T_i - \bar{T}}{S_T} \quad (1)$$

$$CI_i = NPA_i - NTA_i, \quad (2)$$

where P (mm) and T ($^{\circ}\text{C}$) represent the monthly spatially averaged total precipitation and 2m air temperature, respectively, for time period i . The overbar represents the long-term average, which is calculated from 1981–2019, and S denotes the standard deviation. Negative CI values represent a warm/dry climate and positive values represent a cool/wet climate. The more negative or positive a value is, the more extreme the given environment.

Results from annual and seasonal climate index analyses are compared to the total area burned in extreme forest fire seasons, where the area burned during a year is $>5\%$ of a watershed's area. For seasonal analysis, we define the seasons as follows: winter encompasses January, February and December of the previous year, spring encompasses March, April and May, summer encompasses June, July, and August, and fall encompasses September, October, and November. Annual analysis occurs between 1 January and 31 December of a given year and analysis of the water year occurs from 1 October of the previous year through 31 September.

4.2 | Climate suitability classes (CSCs)

To understand how climate affects MPB outbreaks, we use CSCs, first introduced by Safranyik et al. (1975) and more recently used in Carroll et al. (2004) and Taylor et al. (2006), to determine climate suitability for MPB across the watersheds. Safranyik et al. (1975) explore MPB beetle outbreaks and associated climate throughout southern and central BC from 1950–1971, outlining four climate criteria that lead to MPB outbreaks based on air temperature and precipitation. A flow chart of the methods can be found in the supplemental material (Figure S2 in Data S1).

We calculate CSCs at the ERA5-land grid cell scale ($\sim 9 \text{ km}$) to determine the percentage of each class within a watershed's boundary. We first calculate five different variables for each year between 1981–2019, shown in Table 3. We also calculate the long-term average for total precipitation from April through June. Results are then compared to the climate criteria, C1 through C4, from Safranyik et al. (1975) (Table 4). Hourly 2-m air temperature and precipitation from the ERA5-land product are used in all calculations pertaining to CSCs, with daily minimum air temperature used to define criterion C2. Due to year partitions for constraints C1 through C4, annual calculations are performed from August through July of the following year.

After determining if C1 through C4 are met each year, the grid cell's timeseries is split into subsets containing x years. The number of consecutive years where all criteria are met in each subset is counted,

TABLE 3 Variables used to calculate the climate suitability class (CSC) index for MPB

Variable	Description
DD _{gs}	Degree-days above 5.5°C from August 1st to the end of the growing season
DD _y	Degree-days above 5.5°C from August 1st to July 31st
Day _w	Number of days in a year where minimum daily winter temperature < −40°C
T _A	Average maximum August temperatures
P _{sp}	Total precipitation April through June

TABLE 4 Constraints for mountain pine beetle CSC analysis computed at the grid cell scale

Constraint	Description
C1	DD _{gs} > 305 and DD _y > 833
C2	Day _w < 2
C3	T _A ≥ 18.3°C
C4	P _{sp} < P _{sp} long term average

with the sum equating to t . The metric f is then calculated for each grid cell over the subset period, which is converted to a given classification via Table 5.

$$f = \frac{t}{X} \sqrt{X_1 X_2}, \quad (3)$$

where X_1 and X_2 are scaling indices, based on the coefficient of variation (CV) in growing season precipitation and the average index of aridity (AR, in mm) (Table 5). The average growing season for each watershed extends from the last day in spring where the average minimum air temperature dips below −1°C to the first day in the fall where the minimum air temperature falls below −1°C for two consecutive days.

$$AR = \sum_{i=1}^{12} \text{Max}[0, PET_i - P_i], \quad (4)$$

where i is the month of a given year, P (mm) is the total precipitation, and PET (mm) is the potential evapotranspiration (Ung et al., 2001).

Daily PET is calculated using Hamon's (1963) method, and is summed to generate monthly values:

$$PET_i = 0.1651 \times D \times V_d \times K \quad (5)$$

$$V_d = 216.7 \left(\frac{V_s}{273.3 + T} \right); V_s = 6.108 \left(e^{17.269 \left(\frac{T}{T - 273.3} \right)} \right), \quad (6)$$

where D is the time from sunrise to sunset in multiples of 12 hr, V_d (g m^{−3}) is saturated vapour density at the mean daily temperature, T (°C), and V_s (hPa) is the saturated vapour pressure. K is a unitless constant that most commonly falls between 1.2 and 1.4 (Zhou et al., 2015), thus a range of PET values are calculated using those limits, with the average PET being used in the aridity calculations. We selected Hamon's method for calculating PET as it provides a range of values based on readily available variables, such as air temperature, and allows us to be consistent with Zhou et al. (2015) for result comparison (see Section 6.2) (Hamon, 1963). While several other methods can be used to obtain PET , Hamon's PET is both reliable and easily calculable (Lu, Sun, McNulty, & Amatya, 2005).

4.3 | Trend analysis and regression

Both the Mann–Kendall (MK) test (Kendall, 1975; Mann, 1945) and linear regression are used in the analysis process to understand the relationship between climate, forest disturbance, and the hydrologic regime. The MK test is a non-parametric trend test that makes no assumptions about the sample distribution of the dataset and is insensitive to outliers. The MK test requires data independence, thus autocorrelation in the dataset must be removed prior to testing. We tested both annual and seasonal climate indices for autocorrelation finding nearly all lag-1 values fall below the 99th percentile band of ±0.31, suggesting there is no autocorrelation within the climate index time series (Table S1 in Data S1), thus no pre-whitening is performed. The statistical p -value defines the significance of MK test results, with p -value ≤ .10 being significant.

Linear regression is used to assess changes in runoff due to forest disturbance events, with significant results occurring where p -value ≤ .10. While there are many other statistical tests that can be used to detect changes in runoff pre- and post-forest disturbance, the recency of events creates small sample sizes during post-forest disturbance periods. Thus, the majority of statistical analysis methods are not compatible with our dataset.

CV	X ₁	AR [mm]	X ₂	f	CSC
CV < 0.30	0.2	AR < 0	0.2	$f = 0$	Very low
0.30 ≤ CV < 0.35	0.4	0 ≤ AR < 102	0.4	0 < f ≤ 0.05	Low
0.35 ≤ CV < 0.40	0.7	102 ≤ AR < 203	0.7	0.05 < f ≤ 0.15	Moderate
0.40 ≤ CV < 0.45	0.9	203 ≤ AR < 305	0.9	0.15 < f ≤ 0.35	High
0.45 ≤ CV	1.0	305 ≤ AR	1.0	0.35 < f	Extreme

TABLE 5 Conversion table for mountain pine beetle CSC analysis: (1) Coefficient of variation (CV) in growing season precipitation to constant X_1 , (2) index of aridity (AR) to constant X_2 , and (3) metric f to CSC levels

5 | RESULTS

5.1 | Climate and forest fires

Using the MK test, we explore climate index (*CI*) trends from 1981–2019. Summer *CI* has the most significant trends (Figure 2), decreasing by over two points in the most southern watersheds throughout the 38-year period (Table 6). The annual climate becomes significantly warmer and drier in the Spius and Mesilinka watersheds, with annual *CI* dropping by 1.7 and 1.4 points, respectively, over the 38-year study period. The majority of watersheds experience non-significant increases in spring climates and non-significant decreases or no change in winter climates (Table 6).

Comparing climate trends to area burned within each watershed for a given year shows that the frequency of large areas burning annually ($>10 \text{ km}^2$) increases as the summer trend crosses into warm/dry

summer climates (i.e., when *CI* becomes negative) (Figure 2), which is post-1997 for all watersheds except the Stellako and Nautley. The anomalous 1983 forest fire event in the Stellako and Nautley occurred at the end of May, prior to summer, due to exceptionally warm/dry spring conditions (Figure 2g,h).

Between 1981–2019, six extreme fire seasons occur, all after 2008, where the cumulative area burned during the year is $>5\%$ of the watershed's area (Table 7). The extreme fire seasons identified consist of one to four summer fire events that make up over 99.8% of the annual area burned within a watershed (Table S2 in Data S1), thus dominated by large summer fire events. The Nation and Spius are the only watersheds that do not experience any extreme fire seasons. When analysing these events in conjunction with seasonal and annual *CI*, extreme fire seasons occur in years experiencing very warm/dry summer conditions, with an average summer *CI* value of -3.0 during the fire year (Figure 3d). The high *p*-value during the summer is due to

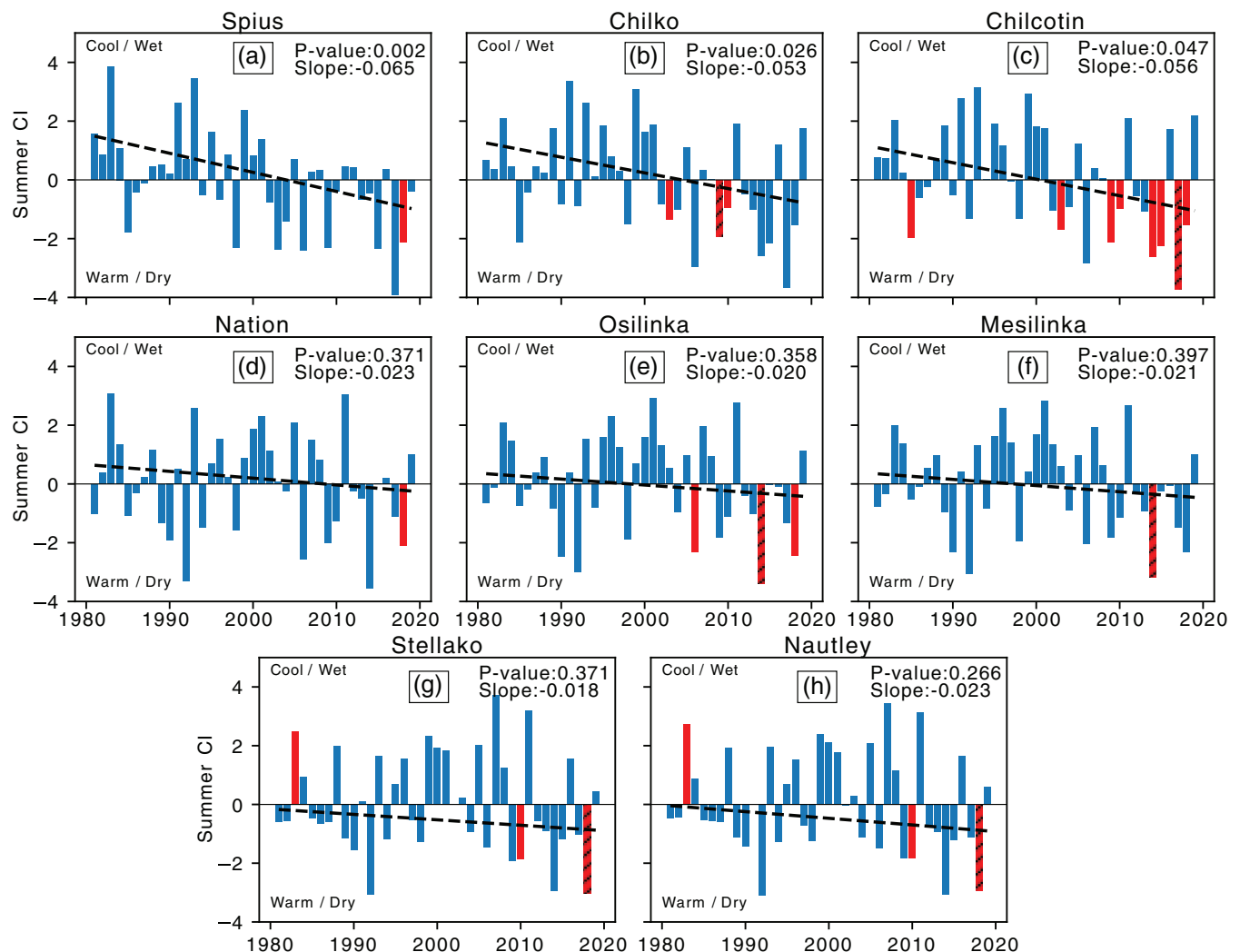


FIGURE 2 Summer climate index from 1981–2019 for each of the eight watersheds. A negative climate index indicates a warm and dry climate and positive values indicate a cool and wet climate relative to normal. Red bars indicate the years when watersheds experienced forest fires that burned $\geq 10 \text{ km}^2$ and bars with horizontal black lines indicate years when $>5\%$ of the watershed burned. The dotted red line is the result of the Mann-Kendall trend test with the *p*-value and slope of the line indicated in the top-right corner of each panel

TABLE 6 Seasonal Mann–Kendall test results for climate index (CI) from 1981–2019, with the total change over the 38-year period shown

Watershed	Annual	Spring	Summer	Fall	Winter
Spilus	−1.70**	−0.59	−2.47***	−0.86	−0.43
Chilko	−0.90	−0.01	−2.03**	0.22	−0.32
Chilcotin	−0.92	0.33	−2.14**	0.30	−0.27
Nation	−0.58	−0.36	−0.88	−0.51	0.53
Osilinka	−1.32	0.08	−0.77	−1.35	0.07
Mesilinka	−1.38*	0.07	−0.80	−1.52*	−0.01
Stellako	−0.08	0.70	−0.70	0.28	0.17
Nautley	0.03	0.56	−0.86	0.21	0.45

Note: Negative values represent the climate getting warmer and drier and positives values represents the climate getting cooler and wetter.

* p -value $\leq .10$, ** p -value $\leq .05$, *** p -value $\leq .01$.

vertical stacking of the data points. The majority of extreme fire seasons follow winters and springs that are slightly cooler/wetter than normal, with the exception of the Stellako and Nautley 2018 fire which experiences a slightly warm/dry spring prior to ignition (Figure 3b,c). Spring CI values have the most significant relationship between climate and area burned, with $p = .15$ (Figure 3c). Furthermore, annual climate is warmer/drier than normal during the year of extreme fire seasons, however there is no distinct linear relationship (Figure 3a).

We also examine the climate in the year leading up to extreme fire seasons, to explore multiyear climate patterns influencing ignition (Figure 3e–i). The majority of extreme fire seasons are preceded by one year of slightly warm/dry summers that are less extreme than the summer climate during the fire year itself, with the lack of a significant linear trend due to vertical stacking of the data (Figure 3h). Increasing relationships between area burned in extreme fire seasons and annual, winter, and fall climates exist in the year prior to forest fires, with the largest fires, Stellako and Nautley 2018, experiencing a cool/wet spring, fall, and winter prior to the extreme fire year and a cool/wet winter during the year of the fires.

5.2 | Forest fire effects on runoff

To understand the effect of extreme forest fire seasons on watershed runoff, we explore the standardized runoff/precipitation (R/P) anomalies over annual and seasonal timescales, pre- and post-fire events. Annual runoff, calculated over the water year, does not significantly change one to two years post-fire, falling within one standard deviation of the mean for all extreme fire seasons (Figure 4a). The Chilcotin 2017 and Stellako and Nautley 2018 fires occur in the same year where summer R/P values exceed the 95th percentile of the 38-year record (Figure 4c). The 95th percentile is also exceeded in spring 1-year post-Chilcotin 2017 fire (Figure 4b) and fall 1-year post-Osilinka and Mesilinka 2014 fire (Figure 4d). Spring R/P during the fire-year of the 2018 Stellako and Nautley events also exceed the 95th percentile, however this is not related to forest fires, as the fires burn in late summer after spring runoff has concluded (Figure 4b). The short R/P record post-Stellako and Nautley 2018 fire prevents long-term analysis of runoff response in these watersheds.

The runoff events that exceed that 95th percentile are compared to normalized precipitation anomalies to explore precipitation regime influences on these events. The large anomalies seen in the Chilcotin in summer 2017 and spring 2018 are not due to forest fires as the same extreme runoff values are experienced in the Chilko (encompassed by the Chilcotin), which was not affected by the 2017 fire season (Figure 5a,d). The high fall runoff values following the 2014 Mesilinka and Osilinka fires appear to be significant outliers of the point cloud distribution (Figure 5e,f) while the high summer runoff values in the Stellako and Nautley occur in extremely dry years, making it difficult to decipher if they are anomalous in relation to the data distribution (Figure 5b,c).

5.3 | Climate effects on MPB outbreaks

MPB outbreaks correspond to a decrease in the percentage of watershed area that has very low MPB suitability (Figure 6). Very low suitability occurs if there are no consecutive years in which constraints

Watershed	Year	Area burned	Fire severity [%]				Burned %
		km ²	Low	Medium	High	Unburned	
Chilko	2009	650.6	3.7	37.1	59.2	0.0	9.5
Chilcotin	2017	2,522.8	31.1	41.1	4.5	23.2	10.1
Osilinka	2014	112.2	4.0	3.6	85.5	6.9	5.4
Mesilinka	2014	505.0	13.3	7.4	73.4	5.9	15.9
Stellako	2018	1,102.0	23.5	34.1	10.7	31.7	18.7
Nautley	2018	1,673.0	24.6	39.0	8.8	27.5	18.5

Note: Area burned is based on the forest fire perimeter (Government of British Columbia, 2019b), with burned percentage representing all burned areas within the fire perimeter (as determined by the Burn Severity dataset (Government of British Columbia, 2020a)). The fire severity indicates the percent of area within the fire perimeter that is categorized as unburned, low, medium, high severity. All extreme forest fire seasons occur during the summer of a given year.

TABLE 7 Summary of extreme forest fires seasons that burned >5% of a watershed's area from 1981–2019

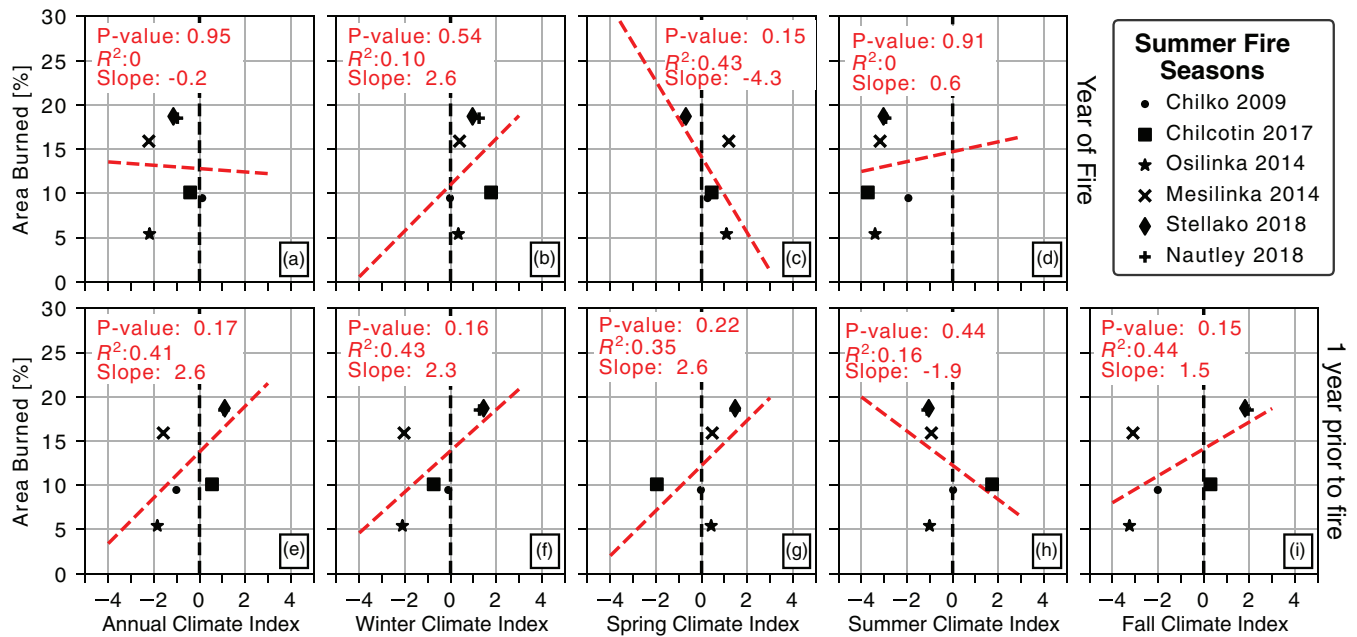


FIGURE 3 Linear regressions between the annual (a,f) and seasonal (b–e, g–j) climate indices for all forest fires that burned >5% of the watershed during the year of the fire (a–e) and the year prior to the fire (f–j). Each symbol represents an extreme fire season in a given watershed with the red line representing the linear regression. The R^2 value, p-values, and slope of the regression are printed in each panel

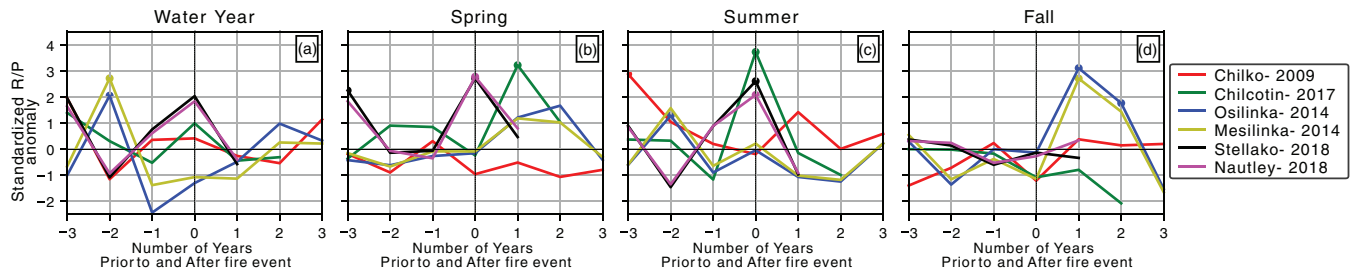


FIGURE 4 Water year and seasonal standardized runoff/precipitation (R/P) anomalies pre- and post-extreme fire seasons. On the x-axis, zero represents the year of the extreme fire season, with negative values denoting the years before the forest fires and positive values denoting the years after the forest fires. Points represent years where standardized R/P anomaly is $\geq 95^{\text{th}}$ percentile of the 38-year record

for the CSC are met. Osilinka and Mesilinka, which lie outside the model calibration region from Safranyik et al. (1975), consistently do not meet climate criterion C1 (Table 4), thus t in Equation 3 is always zero, causing very low suitability in these watersheds (Figure 6e,f). Low and extreme MPB CSCs (Table 5) do not occur in our dataset, thus are absent in our results.

CSC is calculated over 5-year (Figure S3 in Data S1), 10-year (Figure 6), and 15-year timespans (Figure S4 in Data S1). In the 10-year analysis, all watersheds except Osilinka and Mesilinka experience a dip in very low suitability in the 2001–2010 period, corresponding with an increase of medium and high MPB suitability (Figure 6). Four of the eight watersheds (Spilus, Chilcotin, Stellako, and Nautley) are $\geq 50\%$ covered in medium to high suitability from 2001–2010, with the Nation and Chilko covered by $\geq 20\%$ of medium and high suitability. This increase in MPB suitability across the

majority of watersheds corresponds to the years in which MPB outbreaks are most severe. The Nation and Spilus watersheds experience a decrease in low suitability 10 years prior to the most severe MPB outbreaks, from 1991–2000, with very low suitability decreasing by nearly 40%.

In the 5-year and 15-year analysis (Figures S3 and S4 in Data S1), the same general patterns exist with decreasing very low suitability during periods of severe MPB outbreak within the majority of watersheds. The 5-year analysis in the Chilko, Chilcotin, and Nation watersheds (Figure S3 in Data S1) shows sharp decreases in very low suitability ~ 5 years after the start of MPB outbreaks, while in the Stellako and Nautley sharp decreases correspond with the beginning of outbreaks. The 15-year analysis shows a steady decrease in very low suitability over time, except in the Nation where the second 15-year period experiences a slight increase (Figure S4 in Data S1).

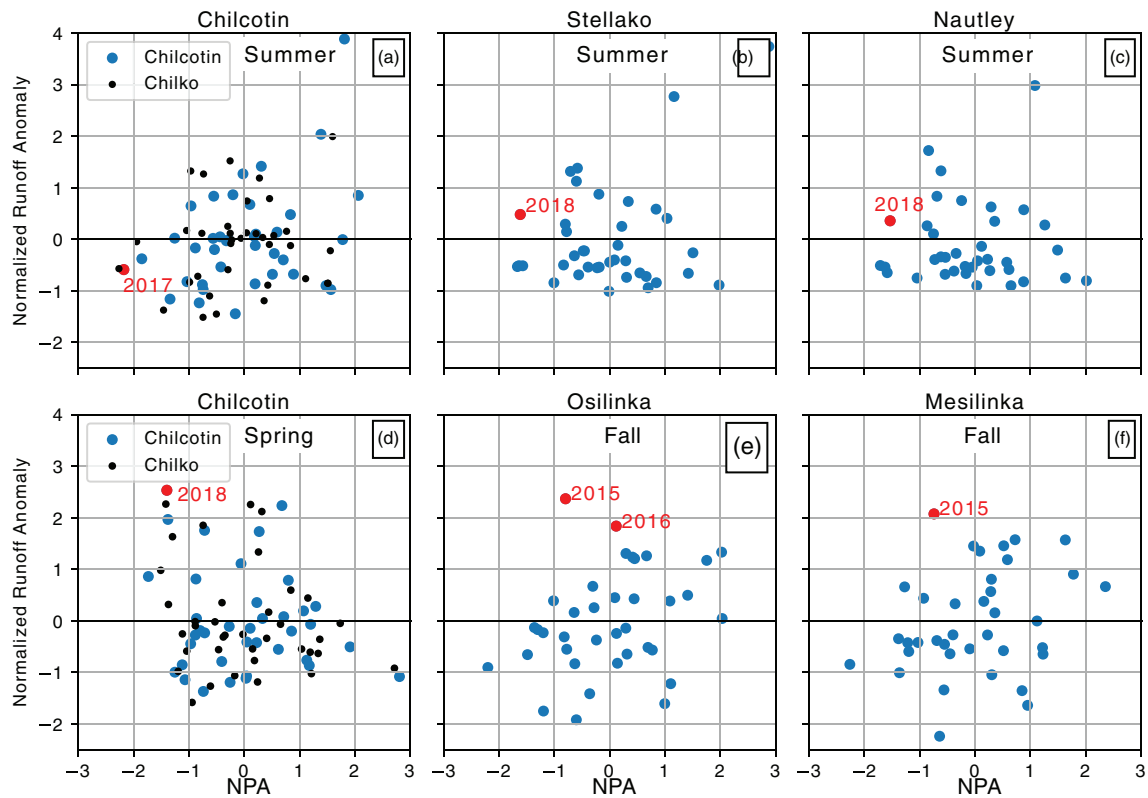


FIGURE 5 Comparison of normalized precipitation anomalies (NPAs) and normalized runoff anomalies. Each panel represents a different season and watershed in which R/P is $\geq 95^{\text{th}}$ percentile, with the specific extreme runoff event highlighted in red (see Figure 4)

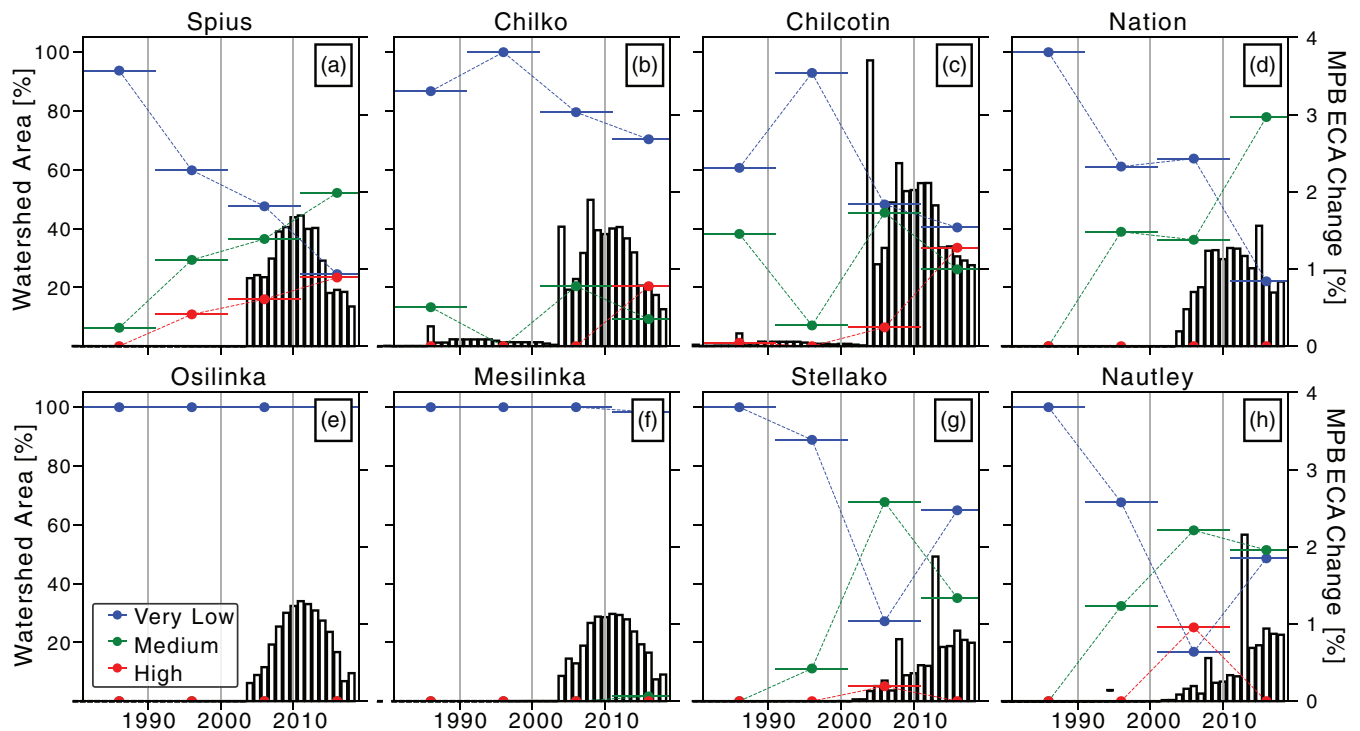


FIGURE 6 Ten-year climate suitability class (CSC) analysis from 1981–2019 for all eight watersheds, each horizontal bar representing a 10-year period. CSC is the percent of a watershed's area that falls into each category. The bar plot represents the change in mountain pine beetle (MPB) equivalent clear-cut area (ECA) between consecutive years

5.4 | MPB outbreak effects on runoff

Linear regression is performed on water year R/P standardized anomalies before and during MPB outbreaks to analyse runoff changes between the two periods. Analysis is performed before the outbreak (1981 to the start of the outbreak: period B), and during the outbreak (start of the outbreak to 2019: period D) (Figure 7). Little change in the standardized R/P anomalies occur between periods B and D, with seven of the eight watersheds' slopes changing by no more than 0.08, with the relationships in the Chilko, Chilcotin, Stellako, and Nautley being nearly identical between the two periods. The largest slope change occurs in the Nation (period D = 0.07, period B = -0.08) along with the lowest *p*-values, at 0.02 and 0.27 for periods B and D, respectively. *p*-values for the majority of linear regressions performed over the B and D periods show large scatter in R/P values, indicating the lack of significant trends. There are also no large differences between the means of periods B and D, with the smallest mean difference of 0.10 occurring in the Spius and the largest difference of 0.83 in the Nautley.

Subset periods that overlap the before and during datasets were also analysed, where logging and forest fire ECA experienced minimal

change ($\leq 3\%$), to remove any influence on the results from other forest disturbance types. Subsets in the Spius, Nautley, and Stellako timeseries had less than 5 years of record, making linear regression inappropriate for analysis. In the remaining watersheds no significant trends in the subsets were detected, with large amounts of scatter occurring in the datasets. Thus, it is unlikely that other forest disturbance types are influencing the results.

6 | DISCUSSION

6.1 | Interrelationships between climate and forest disturbance

Summer climate during the fire year is an important driver of extreme forest fire seasons within the Fraser and Peace drainage basins. Fuel availability is a dominant driver in forest fire ignition and size, with low moisture levels in soil and leaf litter on the forest floor increasing fuel loading (Wotton, Nock, & Flannigan, 2010). Warm/dry summers decrease the moisture content of these layers, increasing the availability of dry fire fuel (Littell et al., 2009). Fire fuel is further enhanced

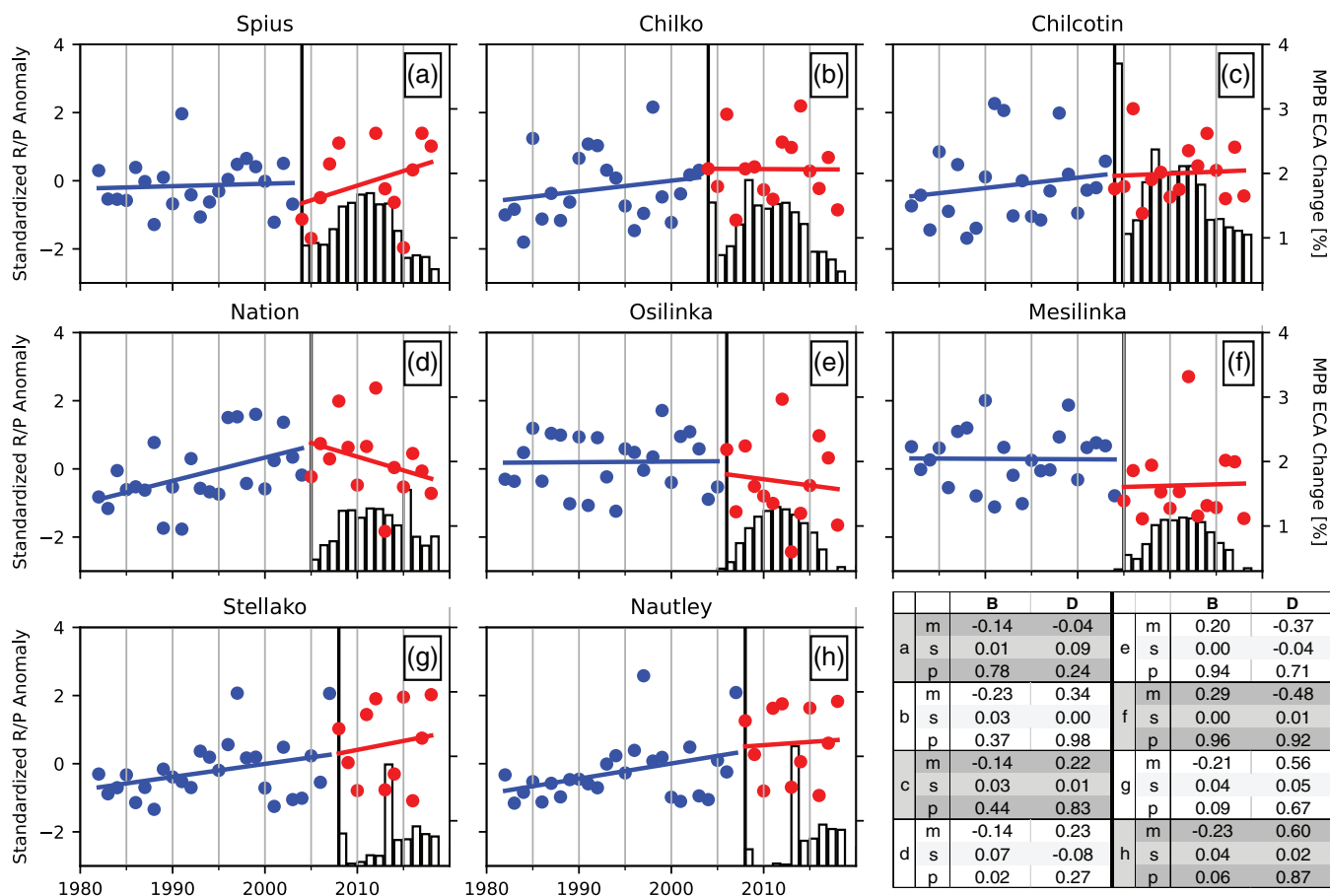


FIGURE 7 Linear regression analysis results before (blue) and during (red) MPB outbreaks of standardized R/P anomalies. Lines represent the linear regression of all years before (B) and during (D) MPB outbreaks. The start of the outbreak is represented by the vertical solid black line and the bar chart represents change in MPB ECA. The table in the lower right-hand corner shows the mean (m), slope (s), and *p*-value (*p*) of the linear regression for each panel and time period

when warm/dry summers are proceeded by cool/wet spring, fall, and winter climates, by increasing vegetation growth (Littell et al., 2009). Increased growth can lead to greater volumes of dry fuel, likely enhancing extreme fire seasons. Long term climate trends from 1981–2019 (Figure 2), show the climate becoming hotter and drier in all eight study sites, which can also contribute to fuel loading through increased tree mortality (Allen et al., 2010) and a decline in total soil moisture, possibly causing increased desiccation of vegetation.

We find that the area burned in extreme fire seasons has an increasing relationship with winter and fall climates in the year prior to the event, with the area burned increasing as these seasonal climates become cooler/wetter. The largest fires, Stellako and Nautley 2018, experience an extremely warm/dry summer preceded by a cool/wet winter, spring, and fall the year prior to the event. Other climatic factors such as wind speed, wind direction, and relative humidity can also affect the intensity, size, and severity of fires (Littell, Peterson, Riley, Liu, & Luce, 2016) by dictating the direction and speed at which the fires move and how long they persist, however this is outside the scope of our work.

A strong relationship between climate and MPB outbreaks in the Fraser and Peace drainage basins also exists. The climate constraints outlined in Safranyik et al. (1975) represent MPB outbreaks well in six of the eight watersheds, with increasing class severity corresponding to the start of MPB outbreaks. However, we find that the model performed poorly in the Osilinka and Mesilinka, the two most northern watersheds of our study sites. The cooler climate and shorter growing season in these northern watersheds cause consistent very low MPB suitability. While the model performed well in watersheds that fell within the Safranyik et al. (1975) model calibration region, the degree day constraint and average maximum August air temperature constraints (C1 and C3 in Table 4) are not good predictors of MPB in more northern watersheds.

The inability of the model to estimate MPB suitability in high latitudes may be due to a more dominant driver of MPB outbreaks in northern regions of BC or overestimation of the model's temperature constraints. There are several environmental factors that influence MPB outbreaks in BC, such as the density and age of host trees and host availability (Alfaro et al., 2015; Aukema et al., 2006). Using the VRI database (Government of British Columbia, 2020c), we find that pine tree species cover 17% of the Osilinka and Mesilinka, with all other watersheds having pine tree extents that exceed 23% (Table 1). While host tree availability is lower in the northern watersheds, MPB ECA is comparable to the southern watersheds, with ~12% MPB ECA occurring by 2018 (Table 1). This suggests that, while there are less host trees in the region, the outbreak severity is not highly dependent on pine tree quantities, thus host tree availability is unlikely to be a dominant driver of outbreaks in the region.

We suggest it is likely that Safranyik et al. (1975) model temperature constraints are not representative of northern BC regions. Currently the model evenly weights the climate constraints; however, some may play a more significant role in predicting MPB suitability than others. For example, winter temperatures have warmed more in northern regions, reducing the number of extremely cold days (<

–40°C) each winter (Vincent, Zhang, Mekis, Wan, & Bush, 2018), which is a primary disruptor of MPB egg development (Safranyik & Linton, 1998). The reduction of extremely cold temperature events may be a more dominant constraint of MPB suitability than other factors, leading to outbreaks in northern regions. Carroll et al. (2004) and Taylor et al. (2006) both note an increase in MPB outbreaks in very low suitability areas, which they attribute to warming climates in higher latitude and altitude regions, thus a revision of the degree day thresholds may also improve the model's suitability predictions. Analysis of in-situ measurements at high latitudes is needed to determine appropriate model improvements, be it weighted climate constraints or lower degree day thresholds.

6.2 | Influence of forest disturbance on river runoff

Effects of forest disturbance on streamflow is usually identified in watersheds where >20% area is affected by forest fire, MPB, and/or logging (Li et al., 2018; Moreno et al., 2020; Stednick, 1996; Wei & Zhang, 2010; Winkler et al., 2017; Zhang & Wei, 2012). Forest fires are much more destructive to the overall composition of a forest than MPB, with fires decreasing soil infiltration rates, creating water repellent soil characteristics, and decreasing vegetation density at all forest levels (Debano, 2000; Moody et al., 2016; Owens et al., 2013). MPB outbreaks induce little change to the understory (Alfaro et al., 2015), which preserves soil properties and vegetation density at ground level. Forest fires also remove large swaths of canopy cover within days or weeks while MPB outbreaks affect only portions of the canopy and, after an attack, needles can remain on pines for several years (Winkler et al., 2014). This likely leads to a muted response in decreasing precipitation interception from MPB outbreaks relative to forest fire events.

The insignificant trends and minimal changes to runoff averages prior to and during MPB events (Figure 7) suggest that there are no large-scale effects of MPB on runoff in the eight watersheds examined. The Chilcotin, which experienced the most severe MPB outbreaks (27.3% MPB ECA), also shows no change in runoff, despite being above the 20% threshold identified in current literature. While MPB outbreaks occur in patches across the Chilcotin, by 2018 many of the patches converge creating large continuous MPB-affected regions. Thus, it is possible that the understory and undisturbed soils mute any response to the loss of large swaths of tree canopy. Biederman et al. (2015) and Slinski et al. (2016) also find a lack of response in watersheds to MPB outbreaks. Biederman et al. (2015) suggest that evenly distributed tree mortality across a region, compared to other forest disturbance types, may lead to increased transpiration and soil uptake by the surrounding understory and trees, which can quickly adapt after MPB attacks.

Effects of forest fire on watershed runoff are difficult to determine without exploring rain events and timing of flow, thus we cannot definitively state if the runoff extremes we detect are due to climatic factors or the fire event itself. When comparing the runoff extremes

to the normalized precipitation anomaly (Figure 5), the fall events post-2014 Osilinka and Mesilinka fires lie significantly outside the point cloud distribution, making these the most likely runoff events related to forest fire activity. It cannot be determined if the other extreme runoff events fall within the general distribution of the historical data, leading to inconclusive results. The largest fire seasons occurred in the Stellako and Nautley in 2018, with fire burning ~18% of the watershed, just below the 20% ECA threshold. The lack of prolonged hydrologic data post-fire events, due to fire recency, prevents us from exploring the long-term runoff effects, however, due to regeneration of vegetation and soil properties over time, we expect the largest impact on runoff to occur in the 1–3 years post-disturbance. Thus, the exploration of short runoff timeseries can still provide valuable insight into hydrologic changes in watersheds.

Our challenges in identifying runoff changes from forest disturbance events may be due to watershed size, with larger watersheds likely having higher water retention capacities than smaller basins (Zhou et al., 2015). Zhou et al. (2015) apply Fuh's equation that explores effects of potential evapotranspiration (*PET*) and precipitation (*P*) on runoff (*R*) by changing the constant *m* which is dependent on watershed characteristics.

$$\frac{R}{P} = \left(1 + \left(\frac{P}{PET} \right)^{-m} \right)^{\frac{1}{m}} - \left(\frac{P}{PET} \right)^{-1}. \quad (7)$$

Zhou et al. (2015) find that watershed characteristics play a more significant role than climate in a watershed's response to forest disturbance if $m < 2$ and a lesser role when $m > 2$. In regions where $P/PET < 1$, water yield is more responsive to changes in watershed characteristics. *PET* estimations vary by calculation method, with higher *PET* values decreasing both *m* and P/PET .

Using Hamon's equation for *PET* (chosen for consistency with Zhou et al., 2015) [Equation (5)] (Hamon, 1963; Lu et al., 2005), we calculate the average *m* value and P/PET ratio for all watersheds from 1981–2019 (Figure S5 in Data S1) to explore how our watersheds fit into the Zhou et al. (2015) framework. The average range of *m* values is greater than two in all watersheds except the Spius and Chilko (Figure S5a in Data S1), and average $P/PET > 1$ (Figure S5b in Data S1), suggesting the majority of our watersheds are not as responsive to forest disturbance changes as climate patterns. This may explain why extreme fire seasons or MPB outbreaks produce no significant runoff changes within our eight basins. More accurate *PET* estimations and a better understanding of water retention capacity and properties would improve our understanding of how runoff responds to forest disturbance in larger watersheds.

7 | CONCLUSION

We find that seasonal air temperature and precipitation are important drivers of extreme fire seasons within the Peace and Fraser drainage basins, with warm/dry summers preceded by multiple cool/wet seasons creating the most suitable environment for large fires. MPB

outbreaks are also dependent on multi-year temperature and precipitation trends to have successful broods and synchronize attacks on tree stands. The CSC model from Safranyik et al. (1975) represents climate drivers in southern watersheds well, however model improvements, such as weighted variables or lower temperature constraints, are needed to better represent northern watersheds in BC.

Effects of forest disturbance on runoff are minimally seen within the eight study sites. The forest understory may help prevent runoff changes from large scale MPB attacks, as limited soil and vegetation changes occur at the forest floor. We expected large fires to cause more prominent changes in runoff due to their effects on soil properties and all levels of forest vegetation, however only two extreme R/P events may be linked directly to extreme forest fire seasons. We suggest the lack of runoff response after large fire seasons may be due to the large watershed size and the percentage of area burned. However, more definitive conclusions on hydrological effects of fires must await further analysis using longer-term data or modelling exercises.

The Fraser and Peace drainage basins would benefit from additional case studies focusing on the effects of forest fire and MPB on streamflow with increased in-situ climate and runoff measurements at a variety of spatial scales. More comparison studies between the hydrologic dynamics in large and small watersheds will improve our knowledge on how different basins will respond to climate change and forest disturbance events in the coming decades. We also suggest increasing research into the effects forest disturbance has on climate as well as the direct effects climate has on the hydrologic system. Many BC communities, economies, and ecosystems rely on the Peace and Fraser drainage basins, making large-scale hydrologic system research important for forecasting potential economic and ecological challenges or opportunities.

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DATA AVAILABILITY STATEMENT

Climate data that support the findings of this study are openly available through Copernicus Climate Change Services at <https://doi.org/10.24381/cds.e2161bac> (Copernicus Climate Change Service (C3S), 2019). Forest disturbance data that support the findings of this study are openly available through the Government of British Columbia at <https://catalogue.data.gov.bc.ca/dataset/harvested-areas-of-bc-consolidated-cutblocks-> (Government of British Columbia, 2020b), <https://catalogue.data.gov.bc.ca/dataset/vri-2019-forest-vegetation-composite-polygons> (Government of British Columbia, 2020c), <https://catalogue.data.gov.bc.ca/dataset/fire-perimeters-historical> (Government of British Columbia, 2019b), and <https://catalogue.data.gov.bc.ca/dataset/>

gov.bc.ca/dataset/fire-burn-severity-historical (Government of British Columbia, 2020a). Hydrologic data that support the findings of this study are openly available through Environment and Climate Change Canada at <https://www.canada.ca/en/environment-climate-change/services/water-overview/quantity/monitoring/survey/data-products-services/national-archive-hydat.html> (Environment and Climate Change Canada, 2020). Code used to manipulate and analyse these data products can be found through GitHub at https://github.com/voremargot/Vore_et_al_2020_HydrologicalProcesses.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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